

JEET BAHADUR THAPA

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Work Authorization: OPT (STEM-extension eligible) | Open to Relocation across USA | Remote, Hybrid, or On-site

SUMMARY

Mechanical Engineer (M.S., May 2026) specializing in CAD design, FEA simulation (ANSYS, Simufact), SLS additive manufacturing, and mechanical testing.

EDUCATION

M.S. Mechanical Engineering, Wright State University — Dayton, OH

Jan 2024 – May 2026

GPA: 3.857 / 4.0

- **Coursework:** Advanced Thermodynamics, Engineering Vibration & Advanced Dynamics, Advanced Mechanics of Solids, Finite Element Method, Additive Manufacturing, Machine Design, Automotive Technology

B.E. Mechanical Engineering, Tribhuvan University — Kathmandu, Nepal

Jan 2011 – Dec 2016

GPA: 3.0 / 4.0

TECHNICAL SKILLS

CAD & Simulation:	SolidWorks, ANSYS (Structural, Modal, Fluent), Simufact Forming, MATLAB, Python
Additive Manufacturing:	Formlabs Fuse 1 (SLS), FDM, Preform, Prusa Slicer; PA12, PA12-GF, PLA, ABS
Mechanical Testing:	Instron UTS (tensile, 3-pt & 4-pt bending, fatigue), MTS 244.51 (ASTM D695 compression), Instron Ceast 9050 (ISO 179-1 Charpy impact), VIC-3D Digital Image Correlation (DIC) full-field strain measurement
Characterization & Metrology:	Keyence VR-6000 optical profilometer, Mitutoyo Vision System, SEM, XRD (Rigaku SmartLab), DSC, TGA, TMA, DMA, X-ray CT, ImageJ
Vibration Analysis:	Spectral Dynamics Bobcat analyzer; RMS, time-domain, and frequency-domain analysis (FRF, transmissibility)
Engineering Methods:	GD&T, DFA, Root Cause Analysis, FMEA, Design of Experiments, SOP development

CERTIFICATIONS

SolidWorks Professional (CSWP), Dassault Systèmes	2025
SolidWorks Associate (CSWA) — 3D Design	2024
SolidWorks Additive Manufacturing Associate (CSWA-AM)	2024
SolidWorks Drawing Tools Professional (CSWPA-DT)	2024
VIC-3D Digital Image Correlation (DIC) Certification, Correlated Solutions	2026
OASIS Rapid Certification on Semiconductors & Integrated Manufacturing	2024

Conference Presentations (Poster), Research & Research

Conference Presentations (Poster) and Research:

- *MS&T 2025, Dayton Engineering & Science Symposium (DESS) 2025, PINO 2026*
 - **Research:** Environmental Effects on Mechanical Behavior of 3D-Printed Glass Fiber-Reinforced PA12 Composites. Studied how build orientation, heat treatment, and moisture change surface integrity and tensile performance in SLS parts.
- **Recipient of the Excellence in Research Poster Presentation Award** — Wright State University Celebration of Research 2026, recognizing outstanding scholarship in designing and presenting a research poster.
 - **Research:** Compressive Behavior and Strain-Rate Sensitivity of SLS-Printed PA12 and PA12-GF Composites Across Multiple Orientations With Microstructural Insights (SEM, TGA, DSC, XRAY-CT, and XRD)

Award: Excellence in Research Poster Presentation Award — Wright State University Celebration of Research 2026, recognizing outstanding scholarship in research poster design and presentation.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Wright State University, Dayton, OH

May 2025 – Present

Additive Manufacturing & Mechanical/Physical Characterization

- Built compression simulation models in ANSYS and Simufact to predict stress and deformation, and validated the numerical results against ASTM D695 experimental data.
- Printed PA12 and PA12-GF specimens on the Formlabs Fuse 1 in multiple build orientations, preparing sample sets for ASTM D638 tensile, ASTM D695 compression, and ISO 179-1 Charpy impact testing.
- Ran ASTM D695 compression tests on the MTS 244.51 at strain rates of 0.01, 1, 3, and 5 (per sec) to measure how strain rate and build orientation change modulus, yield, and stress-strain response.
- Conducted ISO 179-1 Charpy impact tests on the Instron Ceast 9050 to compare orientation-dependent toughness and energy absorption between PA12 and PA12-GF specimens.
- Conditioned ASTM D638 samples in a Watlow F4T chamber and applied controlled heat treatment to quantify moisture and thermal effects on printed part performance.
- Verified specimen dimensions and measured Sa/Sq surface roughness on the Mitutoyo Vision System and Keyence VR-6000, then correlated dimensional accuracy and surface quality to mechanical test results.
- Ran XRD (Rigaku SmartLab) for phase and crystallinity comparison between PA12 powder and printed parts; performed SEM fractography on PA12-GF after cryogenic fracture to evaluate fracture morphology and glass-bead microstructure.
- Used DSC to determine Tm, Tc, and delta-Hm, and TGA to record mass-loss and degradation onset curves for comparing thermal stability of PA12 versus PA12-GF.

Graduate Teaching Assistant | Wright State University, Dayton, OH

Mechanical Vibrations Lab

Aug 2025 – Dec 2025

- Led SDOF vibration experiments (cantilever beam and forced vibration) so students could measure stiffness, damping, natural frequency, and mode shapes.
- Guided FRF and transmissibility analysis using accelerometers, impulse hammers, and rotating unbalance excitation, tying experimental results back to vibration theory.

Mechanical & Material Testing Lab

Jan 2025 – Apr 2025

- Taught labs covering Instron tensile testing, 3-point/4-point elastic beam bending, Charpy impact, heat-treatment effect, and fatigue endurance testing.

Technical Advisor & Sr. Service Engineer | B & K Enterprises, Bhaktapur, Nepal

Dec 2018 – Nov 2023

- Standardized vehicle service SOPs across multiple vehicle classes, which improved shop throughput and saved roughly 225 labor hours per year.
- Led root cause analysis on hydraulic, clutch, and brake system failures; the preventive actions that came out of it cut repeat failures by about 15%.
- Handled warranty and inventory reporting in Microsoft Dynamics NAV, coordinated cross-functional cost and quality targets, and ran emissions compliance inspections using exhaust gas analyzers (gasoline) and opacity meters (diesel).

Service Advisor | Sipradi Trading Pvt. Ltd. (TATA Motors), Nepal

Jul 2016 – Dec 2018

- Improved service turnover by about 20% by tightening the diagnostic workflow (DTC reads, live data, actuator tests, wiring checks via TATA Diagnostics Software) and introducing structured job cards.
- Deployed the Hunter Hawkeye Elite 3D alignment system to get camber, caster, and toe within spec with post-service verification, and trained technicians on updated OEM procedures after returning from TATA Motors HQ programs.

PROJECTS

AGARD 445.6 Wing: FEA for Vibration & Flutter Prediction

Oct 2024 – Nov 2024

- Modeled the AGARD 445.6 wing in SolidWorks, meshed and solved it in ANSYS, and identified the first five modal frequencies under aerodynamic loading for fatigue-failure prediction.

Compressed-Air Engine Retrofit (2-Stroke Gasoline Engine)

Nov 2014 – Nov 2015

- Retrofitted a 2-stroke gasoline engine into a zero-emission compressed-air engine in SolidWorks. Got 0.89 HP at 1886 RPM with 3.11% efficiency, backed by thermodynamic simulation in ANSYS.

Metal Forming Simulation in Simufact — Geometry & Friction Effects

2025

- Simulated cold metal forming in Simufact to see how height-to-diameter ratio and friction coefficient change material flow, barreling, force-displacement response, and stress-strain behavior, & used the output to optimize forming parameters.

Full project write-ups, CAD renders, and test data are on my portfolio: jeetthapa097.github.io/portfolio.